# MAR / HLT

## Intraday Correlation & Cointegration Analysis

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Tickers: MAR (Marriott International) & HLT (Hilton Worldwide)

Date range: 2026-04-01 – 2026-04-07 (excl. Good Friday Apr 3)

Data source: WRDS TAQ millisecond quotes (taqmsec.cqm_YYYYMMDD)

Returns: 1-min non-overlapping | 20-min overlapping (step = 5 min)

Methods: OLS (HC3 SE) · Engle-Granger · Johansen · PCA/SVD

Generated 2026-05-14

Section 1

# Static Correlations

Full-sample OLS regressions with HC3 heteroskedasticity-robust standard errors, run in both directions for 1-min and 20-min overlapping (step=5) returns.

## Table 1 — Static OLS Results

| | Direction | Return | α | β | SE(β) | $R^2$ | N |
|---|---|---|---|---|---|---|---|
| HLT ~ MAR | HLT ~ MAR | 1min | 0.00034 | 0.02867 | 0.03817 | 0.0004 | 1,945 |
| HLT ~ MAR | HLT ~ MAR | 20min | 0.00027 | 0.00933 | 0.10994 | 0.0001 | 382 |
| MAR ~ HLT | MAR ~ HLT | 1min | 0.00092 | 0.01482 | 0.01969 | 0.0004 | 1,945 |
| MAR ~ HLT | MAR ~ HLT | 20min | 0.00046 | 0.00616 | 0.07176 | 0.0001 | 382 |

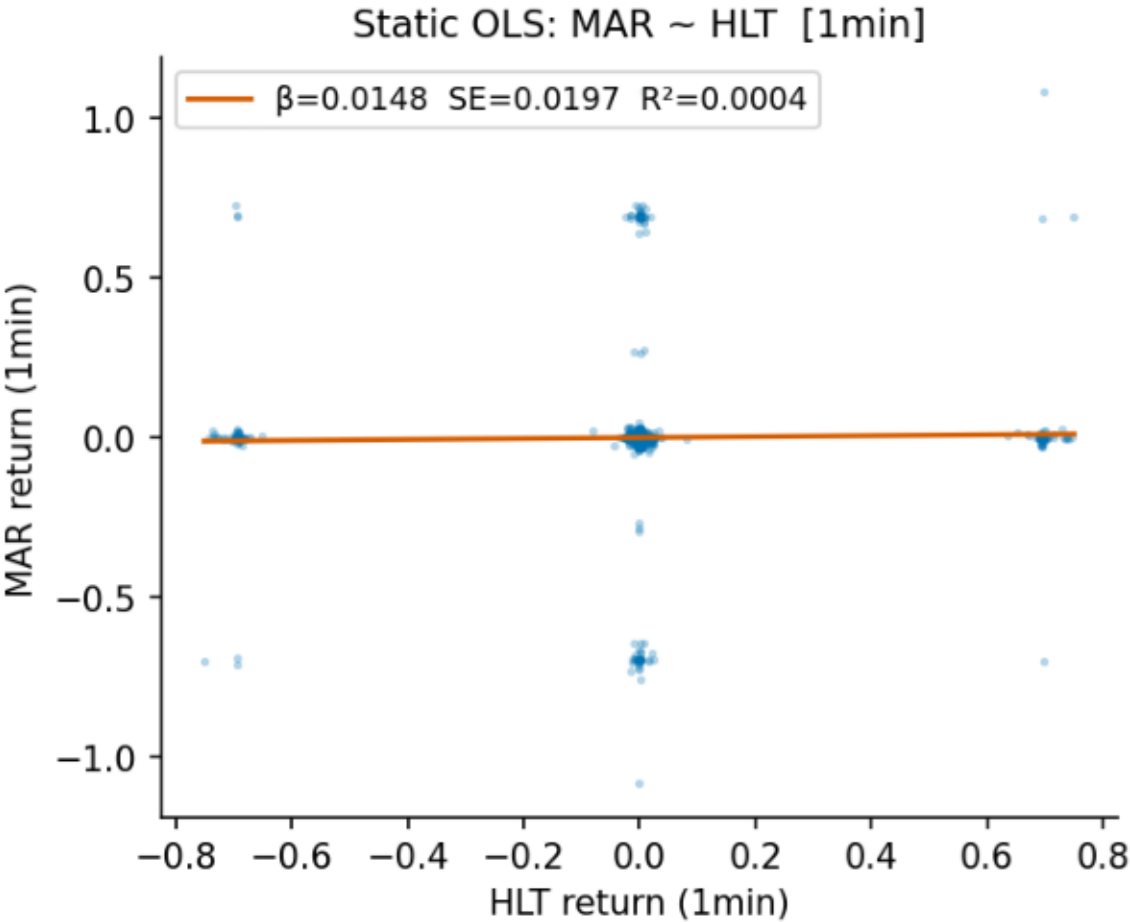


*Figure: MAR ~ HLT, 1min returns — scatter + OLS fit*

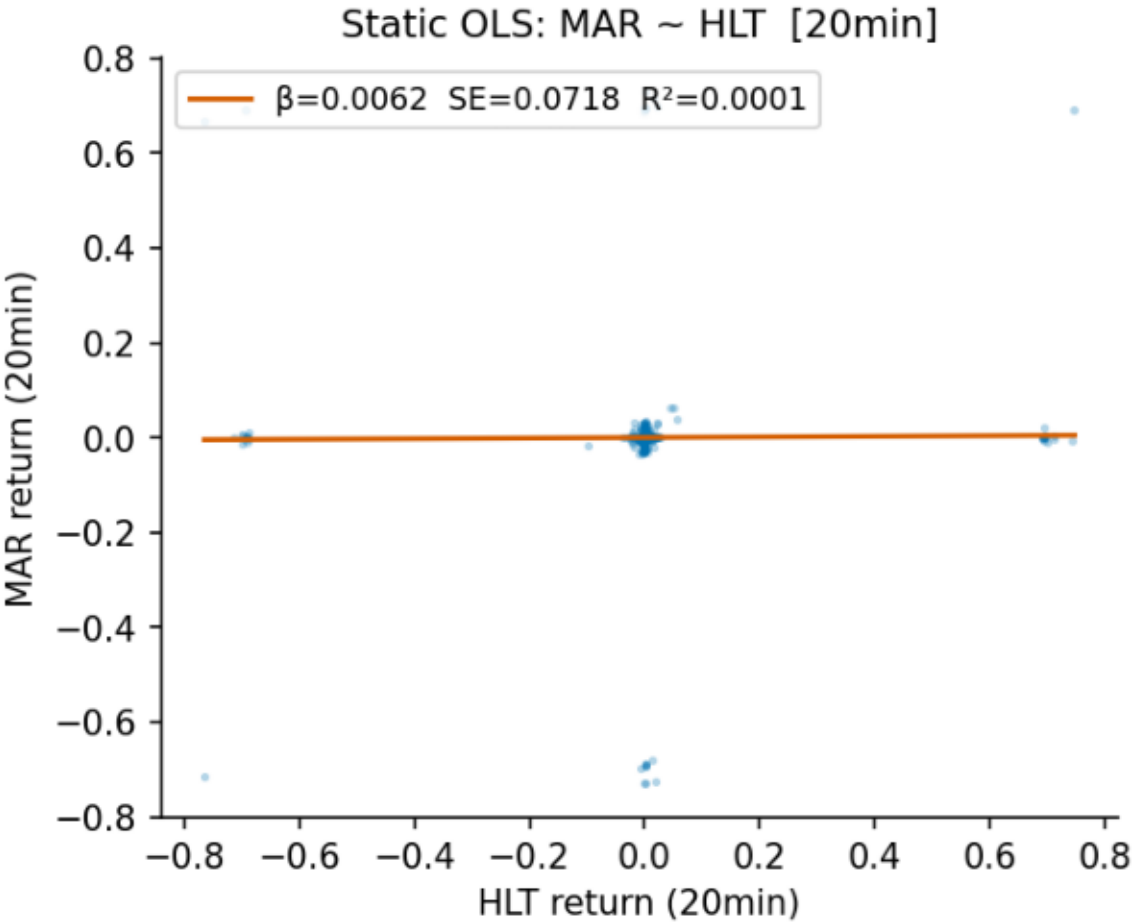


*Figure: MAR ~ HLT, 20min returns — scatter + OLS fit*

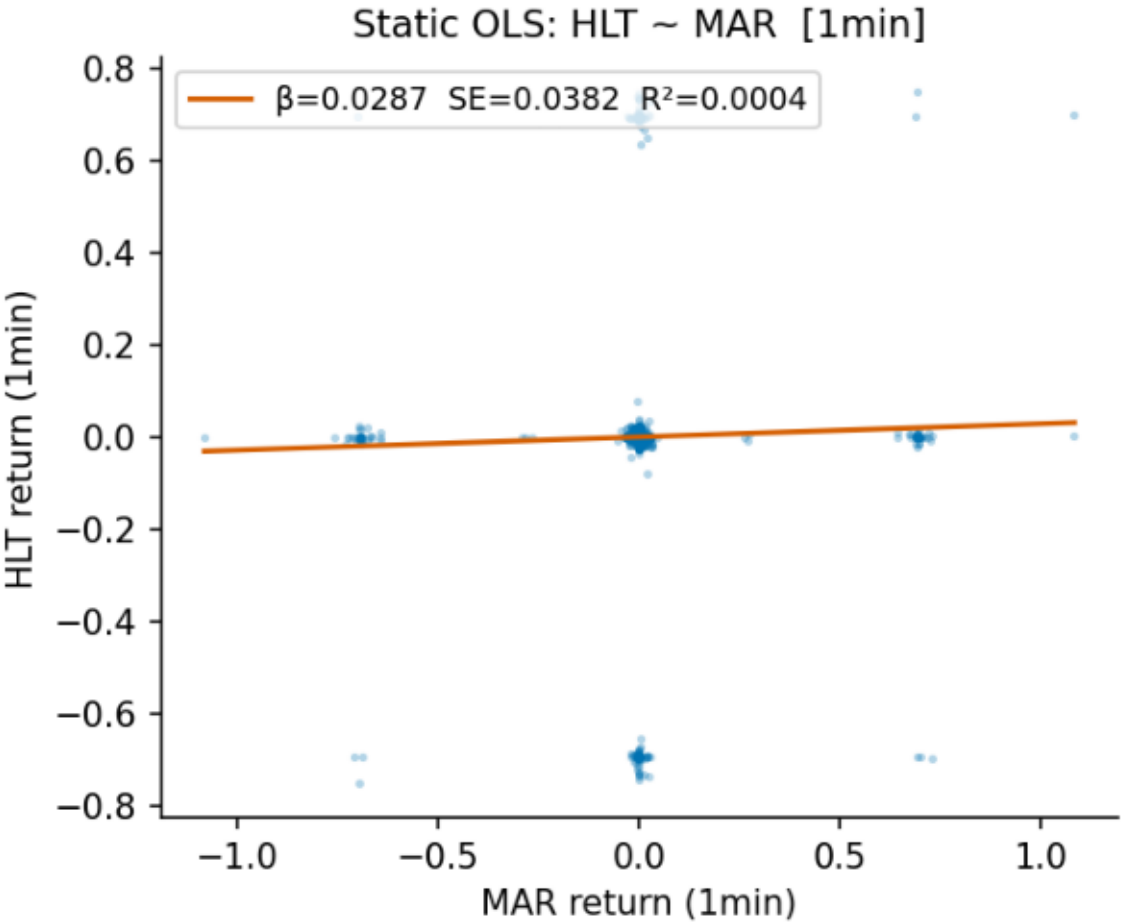


*Figure: HLT ~ MAR, 1min returns — scatter + OLS fit*

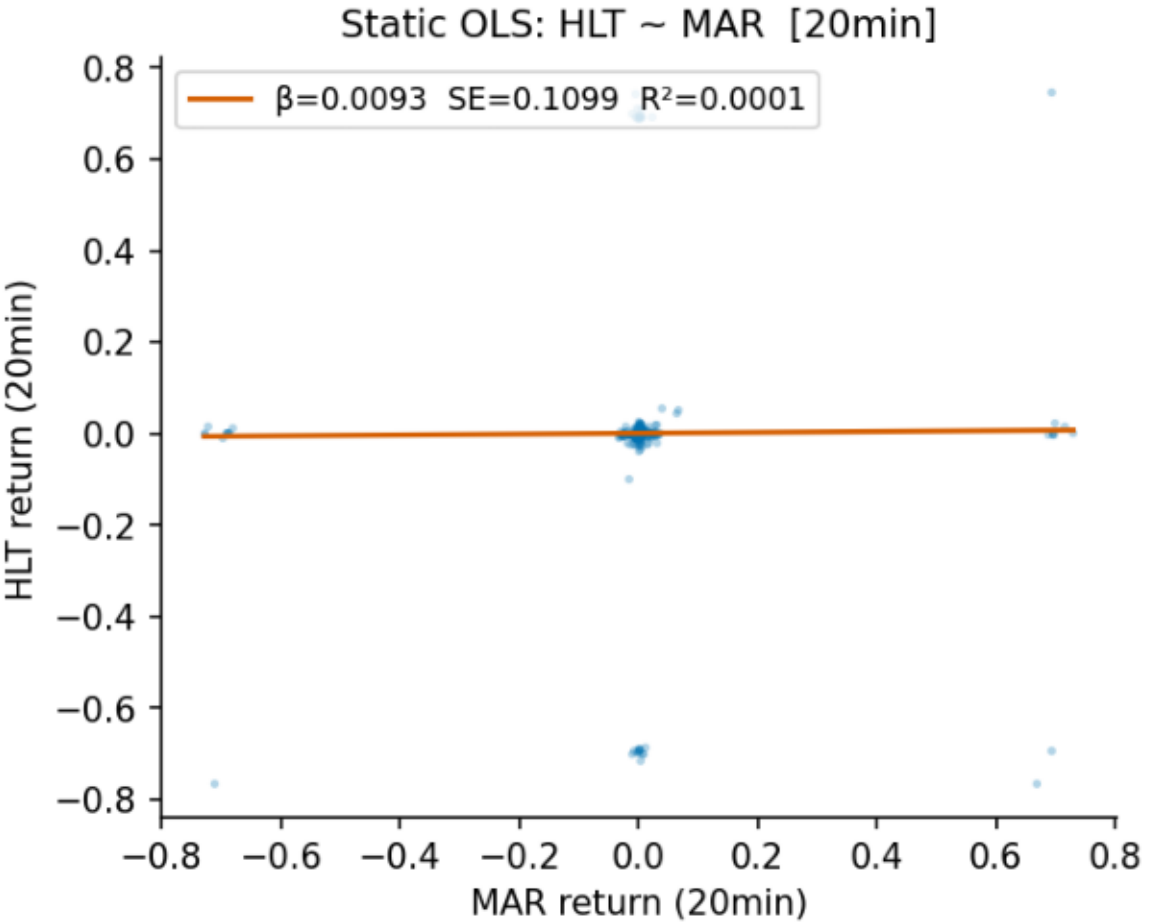


*Figure: HLT ~ MAR, 20min returns — scatter + OLS fit*

Section 2

# Rolling Correlations

OLS regressions on a rolling 1-trading-day window, stepped by ~60 min (1-min returns) or ~12 min (20-min returns). 26 windows per direction per return type.

## Table 2 — Rolling OLS Summary

| | Direction | Return | β mean | β min | β max | SE mean | $R^2$ mean |
|---|---|---|---|---|---|---|---|
| HLT~MAR [1min] | HLT ~ MAR | 1min | 0.0077 | -0.1381 | 0.2240 | 0.0743 | 0.0070 |
| HLT~MAR [20min] | HLT ~ MAR | 20min | 1.0117 | -0.0152 | 7.6129 | 0.6896 | 0.0113 |
| MAR~HLT [1min] | MAR ~ HLT | 1min | 0.0119 | -0.0948 | 0.1527 | 0.0359 | 0.0070 |
| MAR~HLT [20min] | MAR ~ HLT | 20min | 0.0563 | -0.0124 | 0.7039 | 0.0791 | 0.0113 |

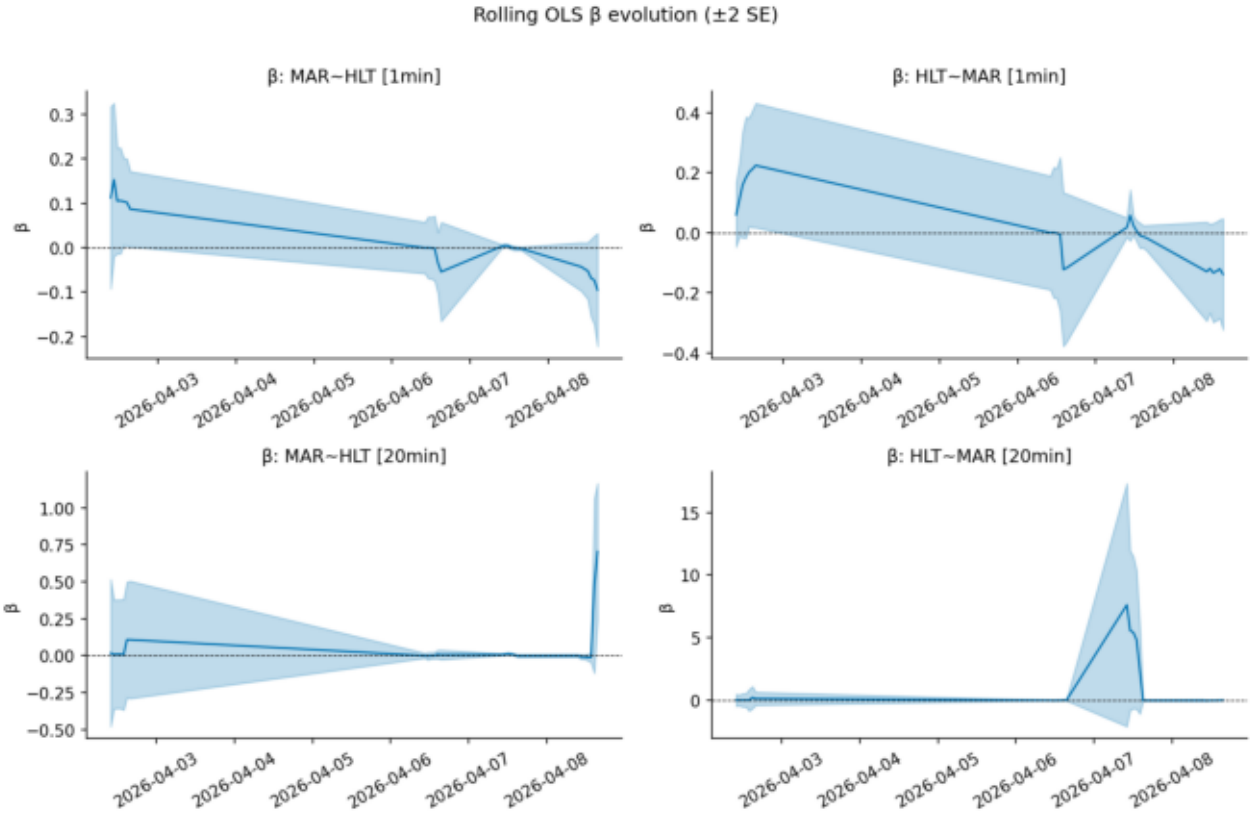


*Figure: Rolling β evolution with ±2 SE bands across all windows*

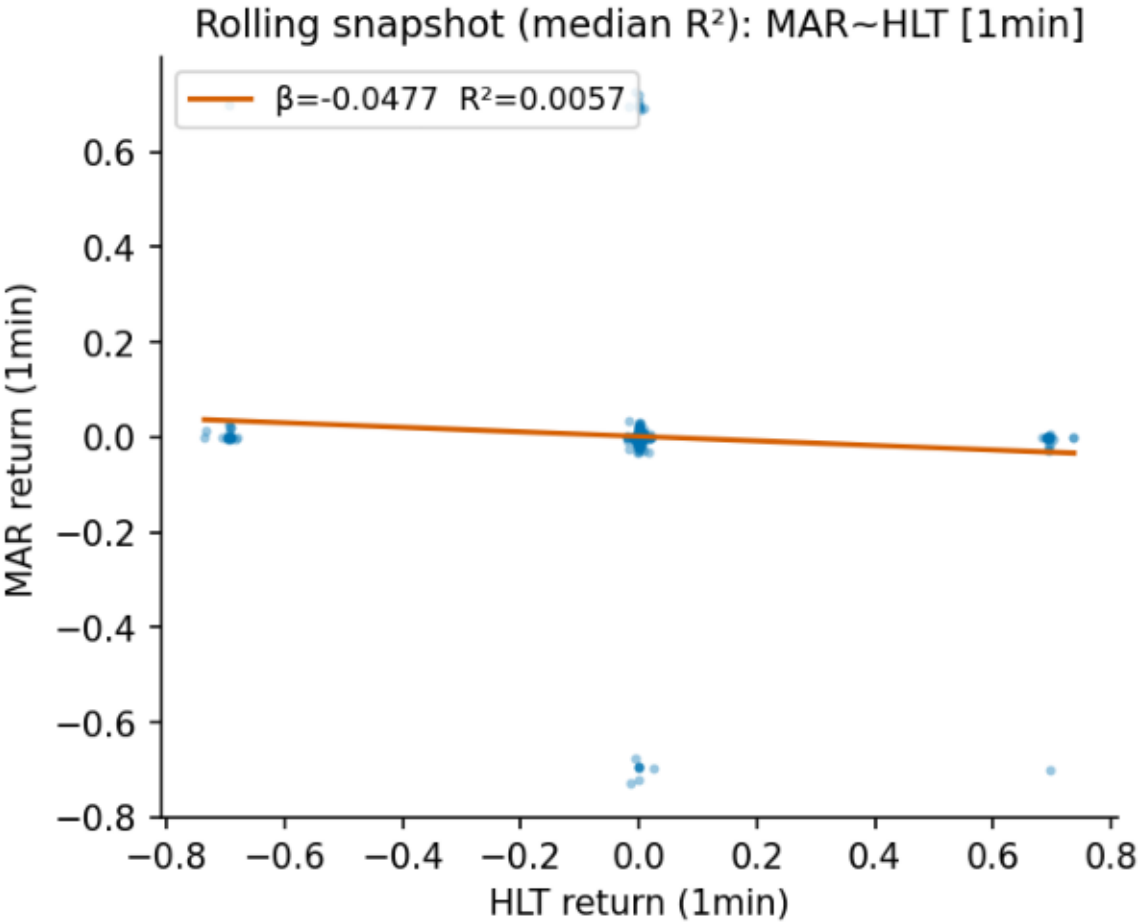


*Figure: Rolling snapshot (median-R² window) — MAR~HLT [1min]*

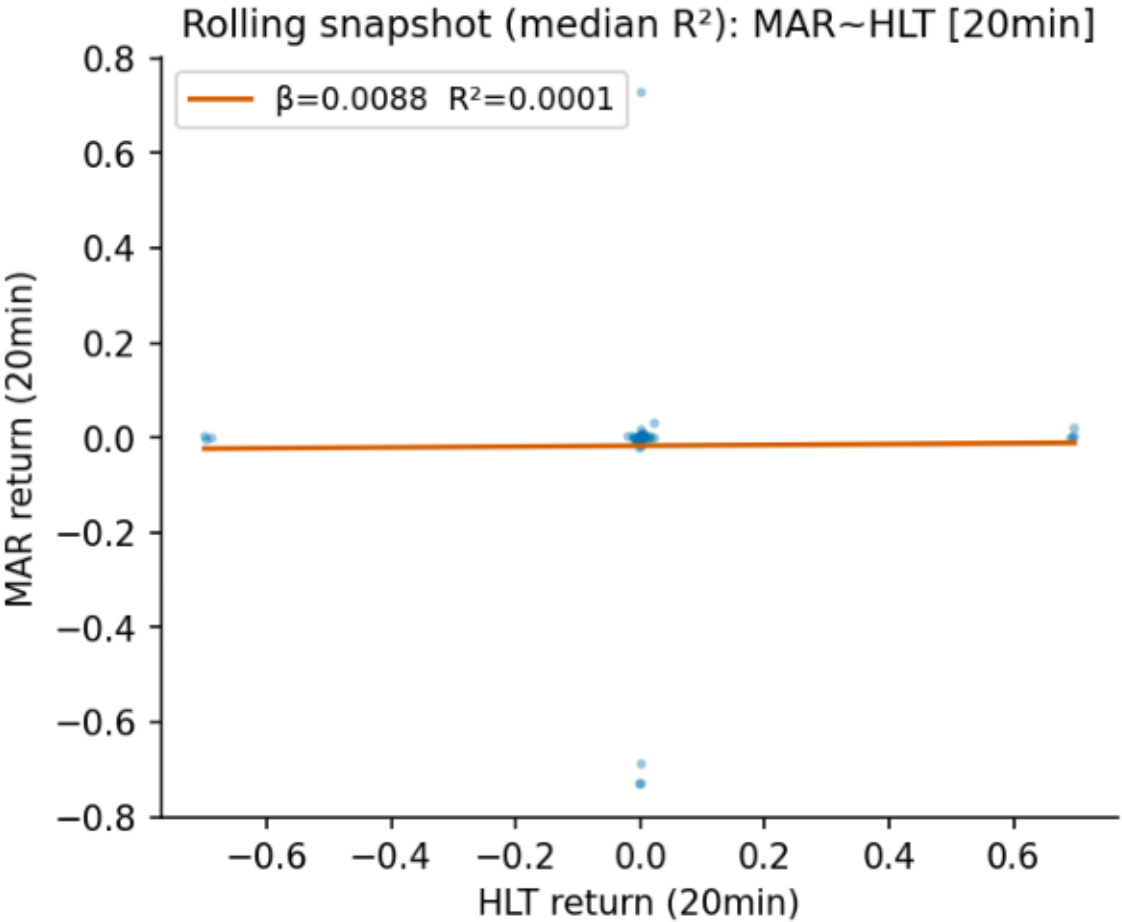


*Figure: Rolling snapshot (median-R² window) — MAR~HLT [20min]*

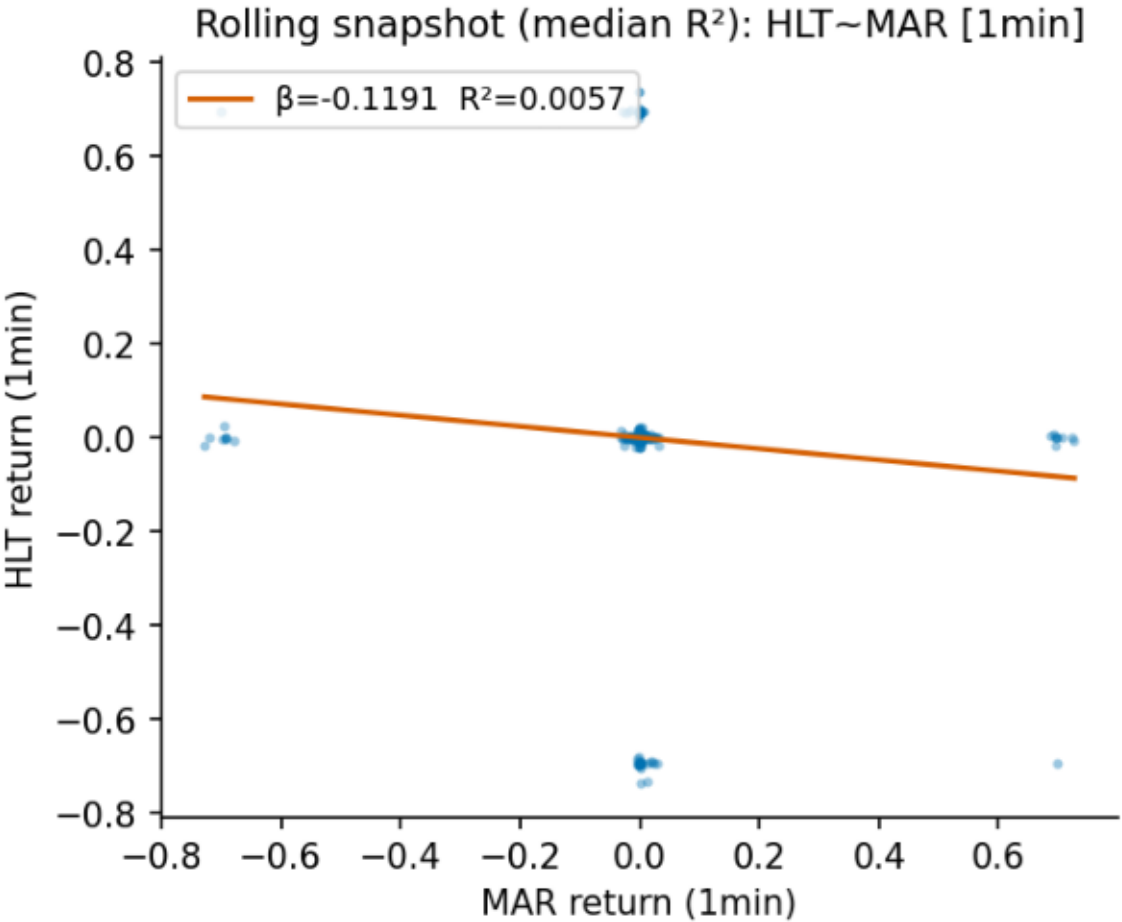


*Figure: Rolling snapshot (median-R² window) — HLT~MAR [1min]*

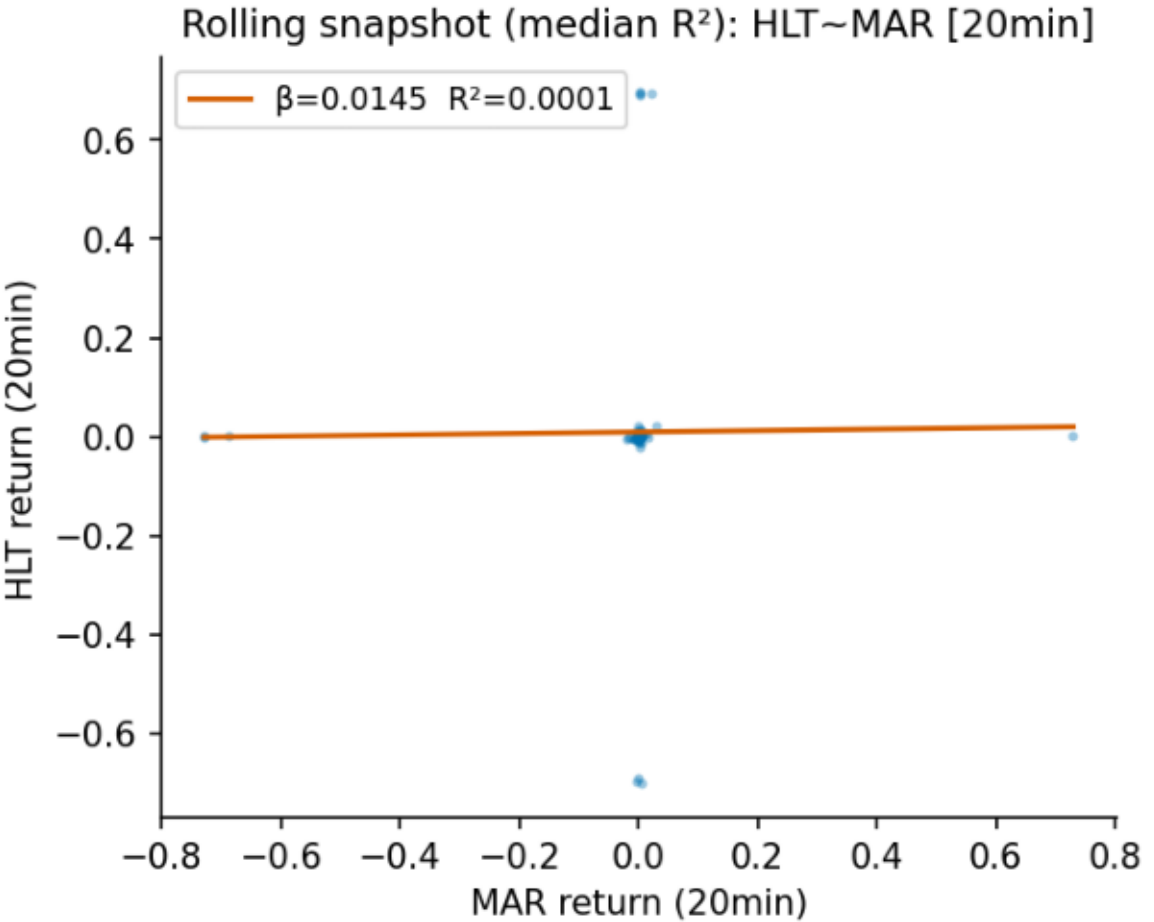


*Figure: Rolling snapshot (median-R² window) — HLT~MAR [20min]*

Section 3

# Autocorrelation

Ljung-Box ACF plots for each ticker. Left panel: 1-min non-overlapping returns (40 lags). Right panel: 20-min overlapping returns, step=5 (20 lags). Bartlett confidence bands shown.

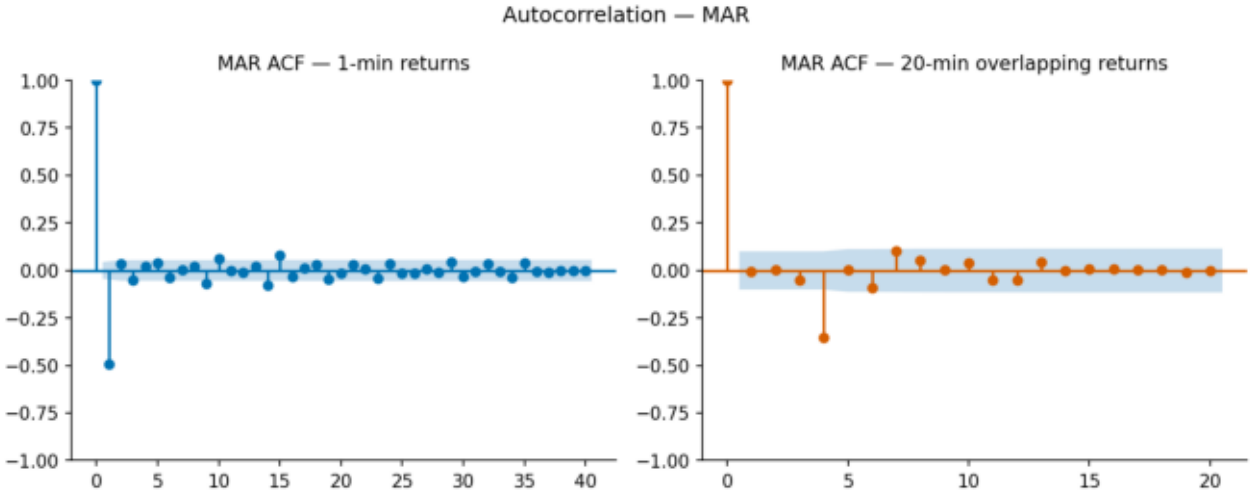


*Figure: ACF — MAR (left: 1-min, right: 20-min overlapping)*

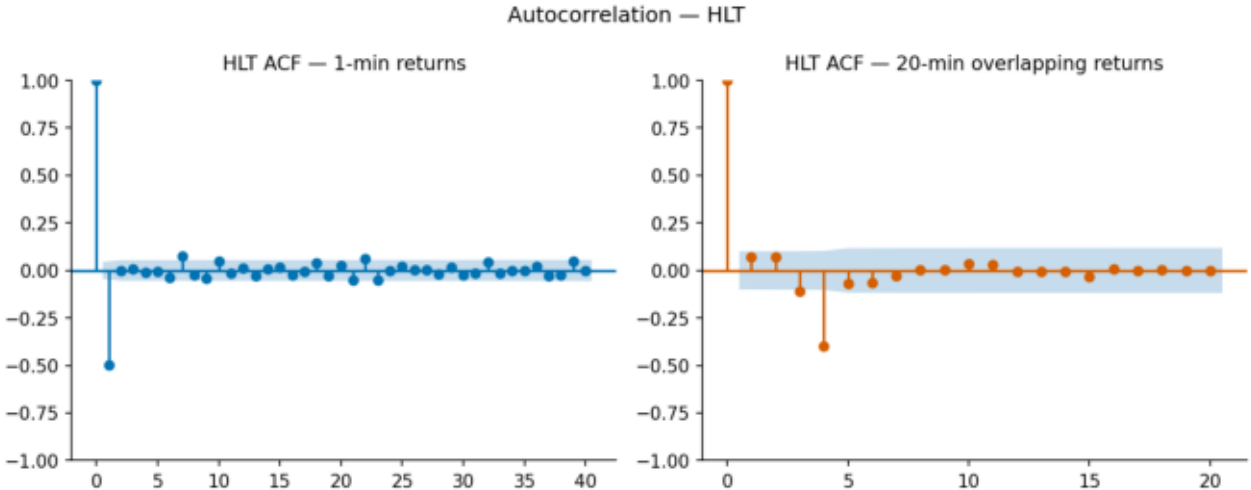


*Figure: ACF — HLT (left: 1-min, right: 20-min overlapping)*

Section 4

# Cointegration

Engle-Granger (EG) and Johansen trace tests on log-price series. Full-span results use all

5 trading days. Daily results show within-day intraday cointegration.

## Table 3 — Full-Span Cointegration

| | Interval | EG stat | EG p-val | Johansen trace r=0 | Johansen trace r≤1 |
|---|---|---|---|---|---|
| 1min | 1min | -7.2873 | 0.0000 | 1212.246 | 558.991 |
| 20min | 20min | -8.6046 | 0.0000 | 220.774 | 91.763 |

## Table 4 — Daily Cointegration

| | Date | EG 1-min stat | EG 1-min p | EG 20-min stat | EG 20-min p | Johansen 1-min | Johansen 20-min |
|---|---|---|---|---|---|---|---|
| 2026-04-01 | 2026-04-01 | -18.887 | 0.0000 | -8.729 | 0.0000 | 256.205 | 58.170 |
| 2026-04-02 | 2026-04-02 | -5.348 | 0.0000 | -1.077 | 0.8873 | 331.106 | 42.893 |
| 2026-04-06 | 2026-04-06 | -20.146 | 0.0000 | -2.760 | 0.1786 | 286.269 | 48.998 |
| 2026-04-07 | 2026-04-07 | -19.571 | 0.0000 | -8.648 | 0.0000 | 297.094 | 66.503 |
| 2026-04-08 | 2026-04-08 | -3.025 | 0.1044 | -8.735 | 0.0000 | 252.176 | 55.108 |

Section 5

# Principal Component Analysis

Per-day SVD calibrated on the first 3 hours of trading (≈179 obs). Two eigenportfolios extracted. Factor z-scores computed with an expanding window from 12:30 onward.

## Table 5 — PCA Eigenvalues by Day

| | Date | EV 1 | EV 2 | EV 1 share (%) |
|---|---|---|---|---|
| 2026-04-01 | 2026-04-01 | 0.069466 | 0.010914 | 86.4% |
| 2026-04-02 | 2026-04-02 | 0.078170 | 0.028379 | 73.4% |
| 2026-04-06 | 2026-04-06 | 0.042117 | 0.000031 | 99.9% |
| 2026-04-07 | 2026-04-07 | 0.094809 | 0.005441 | 94.6% |
| 2026-04-08 | 2026-04-08 | 0.043172 | 0.020760 | 67.5% |

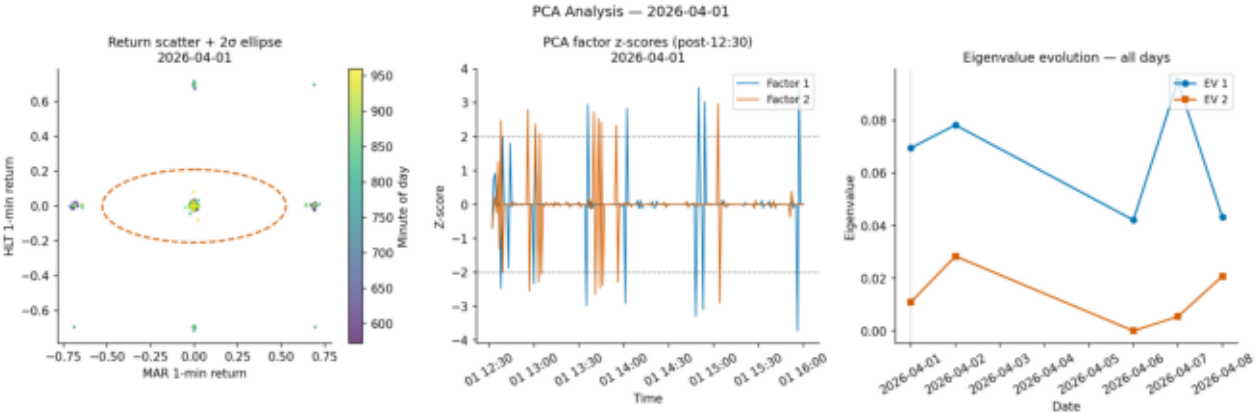


*Figure: PCA — 2026-04-01  (left: scatter+ellipse, centre: z-scores, right: EV evolution)*

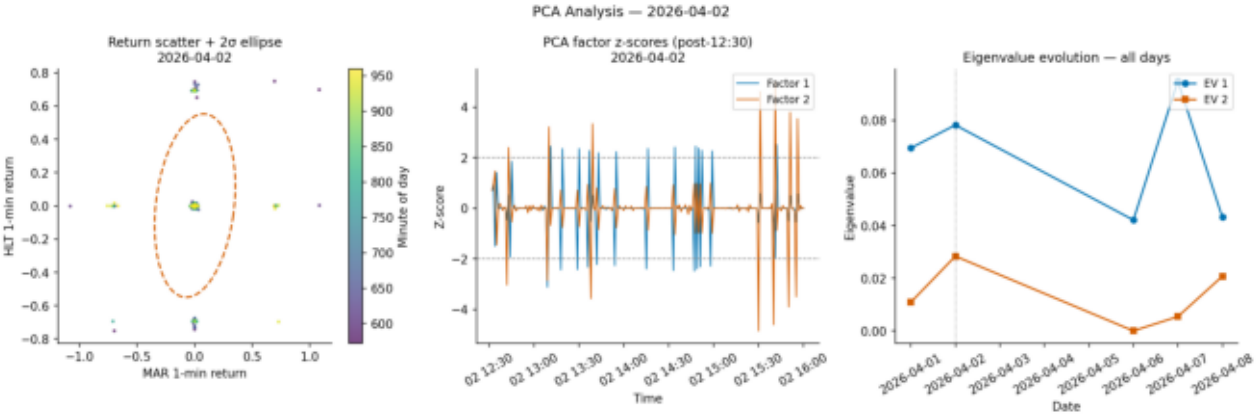


*Figure: PCA — 2026-04-02  (left: scatter+ellipse, centre: z-scores, right: EV evolution)*

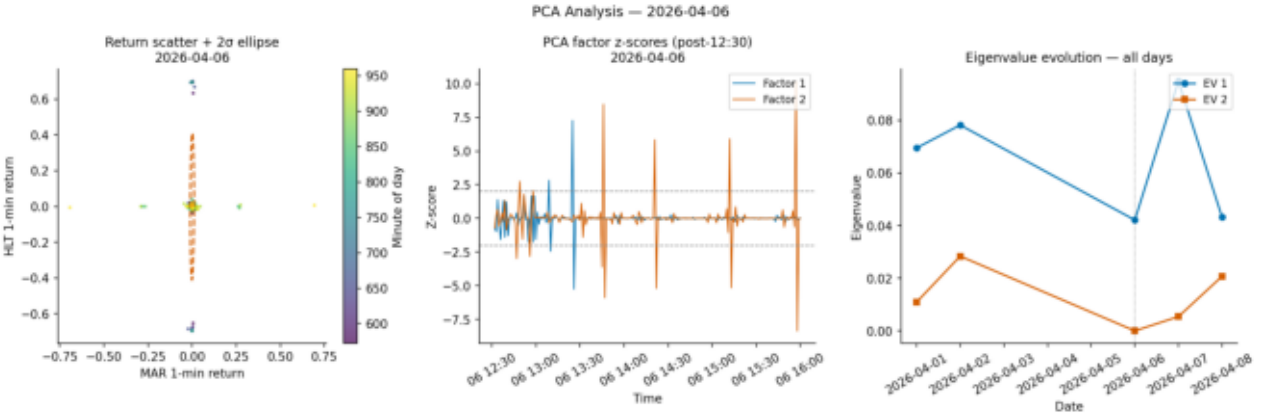


*Figure: PCA — 2026-04-06  (left: scatter+ellipse, centre: z-scores, right: EV evolution)*

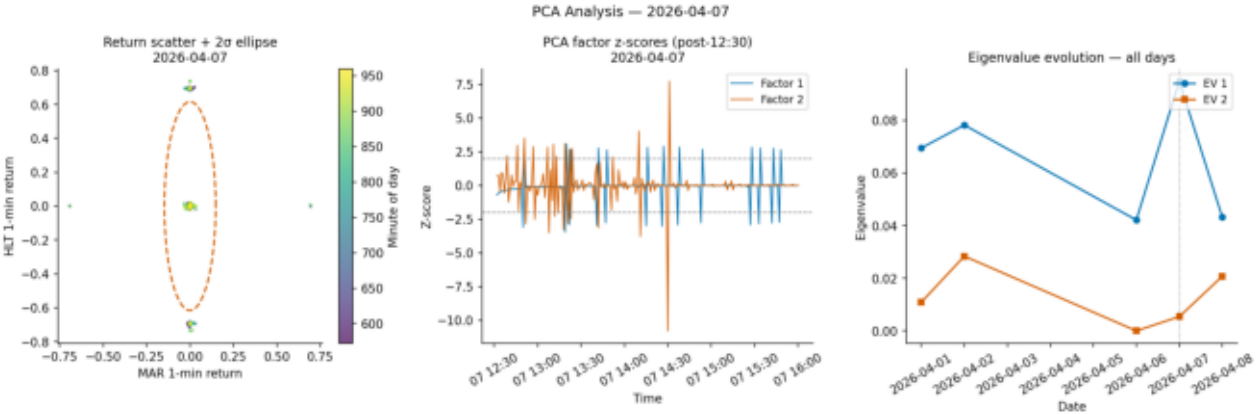


*Figure: PCA — 2026-04-07  (left: scatter+ellipse, centre: z-scores, right: EV evolution)*

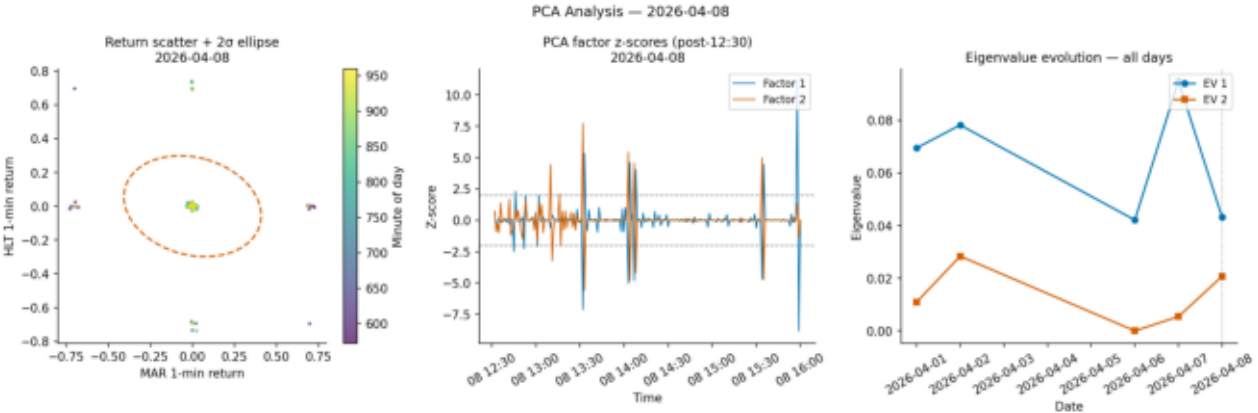


*Figure: PCA — 2026-04-08  (left: scatter+ellipse, centre: z-scores, right: EV evolution)*